

Prajwal Kumar K

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[LinkedIn](#) | [Portfolio](#) | [GitHub](#) | [Codeforces](#) | [CodeChef](#) | [LeetCode](#) | [AtCoder](#)

EDUCATION

Indian Institute of Information Technology, Kottayam

Kottayam, Kerala

Bachelor of Technology in Computer Science and Engineering; CGPA: 8.66/10

Aug 2024 – May 2028 (expected)

TECHNICAL SKILLS

Programming: C++, Rust, Python, TypeScript, JavaScript, Java, SQL, Solidity, MATLAB

AI/ML: PyTorch, TensorFlow, scikit-learn, pandas, NumPy, Matplotlib

Research and engineering: Jupyter, Simulink, FastAPI, Next.js

Research interests: Algorithms and systems, game theory, machine learning and NLP, AI safety

Languages: English (Fluent), German, Hindi, Kannada

EXPERIENCE

Coders Club Coordinator

Jan 2025 – Present

IIIT Kottayam

Kottayam, Kerala

- Organized contests for 80+ participants, curated problems, and configured Codeforces Gym rounds.

Openverse Technical Lead (Open Source Club)

May 2025 – Jul 2026

IIIT Kottayam

Kottayam, Kerala

- Led technical work and pull request reviews for 50+ Openverse contributors.
- Co-built [Algorithmia](#), a visual teaching tool used by professors for DSA and C programming courses.

Full-Stack Web Developer Intern

Dec 2025 – Mar 2026

CIO Mogul (Magazine)

Remote

- Built 3 magazine websites from scratch with a teammate: [CIO Mogul](#), [CIO Mogul India](#), and [CIO Healthcare](#).
- Developed publishing dashboards with Next.js, React, and PostgreSQL on Supabase; automated PDF cover extraction and split double-page spreads for the magazine reader.

PROJECTS

PokeForge | *Python, PyTorch, Gen 9 OU*

[Code](#)

- Prototyped a Pokémon agent evaluation testbed with 2 pinned baselines, FoulPlay and Metamon (in progress).
- Built PyTorch MLP and Transformer policies with legal-action masks and 61 offline contract tests; simulator conformance remains under evaluation.

Bluff | *Python, PyTorch, Bayesian modeling*

[Code](#)

- Designed a 2-player imperfect-information card game to study bluff detection and opponent adaptation.
- Modeled opponents with Bayesian updates and trained a bluff detector; improved draw-aware score by 4.9–5.8 percentage points across 8 synthetic profiles versus a baseline without the detector.

Drishti-XAI | *Retinal imaging, MATLAB, Python*

- Built an explainable retinal-screening prototype in MATLAB and Python; assessed generalization on 2 datasets.
- Measured sensitivity and specificity at a fixed model and threshold: 90.8% and 87.7% on APTOS (n=733), versus 36.3% and 86.5% on Messidor-2 (AUC 0.666).

SiliconSpire | *C++, Rust, Metaheuristics*

[C++](#) | [Rust](#)

- Engineered C++ and Rust optimizers for facility placement using the Quadratic Assignment Problem formulation.
- Combined Grey Wolf Optimization and Tabu Search; matched published optima in 77 of 90 Rust runs (85.6%) versus 34 of 90 C++ runs on 3 small QAPLIB instances.

BattleCP | *Rust, Axum, Tokio, WebSockets*

[Live](#) | [Code](#)

- Developed a multiplayer Battleship game that awards turns for solving Codeforces problems.
- Implemented a Rust WebSocket server with game rules, Codeforces submission checks, and session recovery; recorded 6,922 cumulative page views.

ACHIEVEMENTS AND CERTIFICATIONS

Publication: First-listed author and presenter, “Collaboration-Enabled Energy-Aware Learning Model Exchange (CE-LME) Framework for Edge Computing-Assisted Applications,” IEEE ICICI 2025. [Paper](#).

[Algorithmia](#): selected for and presented at OOSC 3.0, IIT Kanpur (2025); team presentation at T4E 2025, IIT Madras.

Hackathon: Won Best Software Award at IEEE MACE .hack(); 2026 with Team Hogorithm (Anchor).

Programming contests: ICPC 2025 team rank 2,273 (bnb); IICPC 2026 rank 3,352.

Competitive programming: Codeforces Pupil (peak 1221); CodeChef peak 1463; 400+ problems solved.

Cyfrin Updraft (2025): Blockchain Basics, Rust Programming Basics, Solidity Smart Contract Development, and Fundamentals of Zero-Knowledge Proofs.